FLAIR: A Federated Benchmark for ICU Clinical Decision Support Across 12+ Academic Medical Centers Using the Common Longitudinal ICU Format

CLIF Consortium
FLAIR Working Group

June 5, 2026

Abstract

Background. AI clinical decision support (CDSS) tools are proliferating in practice, yet external validation remains concentrated on a handful of public ICU electronic health record datasets. Three problems result: (i) *benchmark saturation*—repeated tuning on the same public test sets inflates reported performance; (ii) a *reproducibility gap*—few publications release runnable code, so retrospective re-execution at a new site is slow or infeasible; and (iii) *incomplete reporting*—most studies omit full TRIPOD+AI items and subgroup fairness metrics that regulators increasingly require. Bilateral data use agreements stretch multi-site validation over years and rarely reach more than a few sites.

Methods. FLAIR is a federated benchmark in which code travels to data, never the reverse. It is anchored on the Common Longitudinal ICU Format (CLIF), a common data element–aligned critical-care schema adopted by 12+ academic medical centers. Five binary classification tasks span outcome, deterioration, and intervention targets (ICU daily mortality, daily discharge to long-term acute care, extubation failure, sepsis, KDIGO stage 2 acute kidney injury). Models run in an immutable container at each site and emit aggregate-only JSON reports (cohort characteristics, full TRIPOD+AI metrics, subgroup fairness gaps, calibration curves) with HIPAA cell suppression. The full FLAIR stack—library, task definitions, container specification, and report generators—is released open source (Apache 2.0) at `github.com/clif-consortium/FLAIR`, and the task catalog is designed to grow: any contributor can add a new benchmark task as a single-file Python module via GitHub pull request.

Results. Across ∼12 consortium sites and ∼500,000 ICU stays, baseline models trained on MIMIC-CLIF show moderate AUROC retention with clinically meaningful calibration drift on external sites **TODO:** populate from federated run.

Conclusions. FLAIR shrinks retrospective multi-site CDSS validation from years to weeks, exposes models to a growing federated cohort that resists public-benchmark saturation, and produces the TRIPOD+AI and fairness evidence package regulators increasingly require—without patient-level data leaving any site. The initial five-task suite is a seed, not a ceiling: the open governance model invites the community to extend the benchmark to new clinical questions over time.

1 Introduction

The bottleneck in clinical decision support (CDS) for the intensive care unit is no longer the supply of models. Predictive algorithms for sepsis, acute kidney injury, extubation readiness, and ICU mortality have been published at a rate the literature struggles to absorb. What the field is missing

is the evidence that any given model behaves acceptably outside the institution that built it. The body of evidence required to support routine clinical use has not kept pace with the proliferation of models, and that gap is, in our view, primarily an *access* gap: representative, multi-institutional data on which to test a model rarely sit within reach of the team that needs them.

The evidence gap is an access gap. External validation of even flagship EHR-based CDS has remained stubbornly narrow. The Epic Sepsis Model, deployed in hundreds of United States hospitals, was first independently evaluated at exactly one site, with discrimination falling to AUC 0.63 against vendor-claimed ranges of 0.76–0.83 (Wong et al., 2021; Habib et al., 2021). Duke's Sepsis Watch, an exemplar academic deployment, took more than six years to be assessed at its first external site (Sendak et al., 2025). The Targeted Real-time Early Warning System (TREWS) reached five sites, all within a single health system (Adams et al., 2022). Mayo Clinic's electrocardiogram-based low-ejection-fraction screen, taken through the FDA pathway, required multi-year coordination to cover roughly four to six external sites (Carter et al., 2025). Each of these efforts required a bilateral data use agreement, shipping de-identified patient data to a coordinating analytic center. None scale to dozens of sites at the pace the rest of the AI ecosystem moves.

Public benchmarks saturate; private benchmarks don't scale. The community has filled some of the gap with open ICU benchmarks (van de Water et al., 2023; Yèche et al., 2021) built on MIMIC, eICU, HiRID, and AUMCdb. These tools are essential for development but inherit the selection biases of the source datasets (Röösli et al., 2022), and the literature now documents how repeated tuning against a fixed public benchmark produces "illusory generalizability" (Recht et al., 2019; Chekroud et al., 2024). Federation-by-data-use- agreement helps but is narrow in scope: the Observational Health Data Sciences and Informatics network federates SQL queries and coefficient sharing for logistic models on a pre-harmonized common data model (Reps et al., 2020); MedPerf and the FeTS Challenge target imaging (Karargyris et al., 2023; Pati et al., 2022). None of these combine modern, longitudinal EHR-based CDS, a clinically meaningful ICU schema, and an extensible task catalog open to the community.

Contribution. We present FLAIR (Federated Learning Assessment for ICU Research), a benchmark framework with three load-bearing properties. First, it operates on the *model-to-data* paradigm pioneered by the DREAM Challenges and applied at scale to EHRs in the EHR DREAM Challenge (Guinney and Saez-Rodriguez, 2018; Bergquist et al., 2024), but extended from one-time competitions to a standing, repeatable infrastructure. Second, it is anchored on the Common Longitudinal ICU Format (CLIF), a critical-care common data element schema adopted by 12+ academic medical centers (Rojas et al., 2025), conceptually parallel to mCODE for oncology (HL7 International, 2024) and complementary to the Medical Event Data Standard (Arnrich et al., 2025). Third, FLAIR is released open source (Apache 2.0, `github.com/clif-consortium/FLAIR`): the task catalog, evaluation protocol, and submission contract are extensible by GitHub pull request, with each new task added as a single Python module under `flair_benchmark/tasks/`. The benchmark co-evolves with clinical need rather than ossifying around a single author group's priors, and the five-task v1 suite below is explicitly a seed for that catalog rather than a fixed scope.

We organize the rest of the manuscript around three commitments: the CLIF-anchored *datasets* that participating sites contribute (§3), the initial five-task *benchmark* suite together with the open contribution protocol for adding more tasks (§4), and the federated submission and evaluation

protocol (§5). A short subsection (§6) then situates FLAIR within the kind of evidence regulator-facing CDS development now requires.

2 Methods

We describe FLAIR as three coupled artifacts: a data layer (CLIF, with MIMIC-CLIF as the public development entry point), a task layer (five benchmark definitions implemented as single-file Python modules), and a protocol layer (containerized model-to-data execution with aggregate-only JSON reporting). Detailed specification is deferred to §3–§5; per-task algorithmic schematics are in the Supplementary Appendix.

Reporting follows the TRIPOD+AI statement (Collins et al., 2024); the FLAIR library auto-populates the checklist into the per-task `tripod_ai.json` report. A Datasheet for Datasets (Gebru et al., 2021) accompanies the public MIMIC-CLIF cohorts, and a Model Card (Mitchell et al., 2019) accompanies the baseline gradient-boosted-tree models.

Institutional review. Each participating consortium site obtained independent institutional review board (IRB) approval for participation in CLIF and for execution of FLAIR benchmark code on its patient data. No patient-level data are transmitted between sites; only aggregate JSON outputs leave each institution. Site governance retains final approval over which submitted models are executed.

Data availability. MIMIC-CLIF is publicly accessible via PhysioNet under the standard MIMIC data use agreement. Consortium data are not redistributable; they are accessible only by submitting code through the FLAIR portal for federated execution at participating sites.

Code availability. The FLAIR library, task definitions, report generators, and the CLIF-WorkBench container specification are released under Apache 2.0 at `github.com/clif-consortium/FLAIR`.

3 Datasets

The Common Longitudinal ICU Format

CLIF is a critical-care minimum dataset that harmonizes ICU electronic health record content into a small set of relational tables with standardized vocabularies for vital sign and laboratory categories, medication routes, and respiratory support modes (Rojas et al., 2025). As of the founding paper, CLIF spanned ∼9 health systems, ∼39 hospitals, and ∼111,000 ICU admissions; the consortium has since grown to 12+ academic medical centers covering ∼2,000 ICU beds. The schema is intentionally narrow — vitals, labs, medications, respiratory support, admission–discharge–transfer, and patient demographics — so that contributing sites can map their EHR exports once and reuse the mapping across studies. In spirit, CLIF is to critical care what mCODE (HL7 International, 2024) is to oncology: a clinically scoped common-data-element layer that makes downstream analytic code portable. CLIF is complementary to broader EHR standards such as the Medical Event Data Standard (Arnrich et al., 2025) and OMOP; ETL adapters between CLIF and these formats are maintained so that adoption does not require schema lock-in.

MIMIC-CLIF as the public development entry point

To support open development, the consortium distributes MIMIC-CLIF: the MIMIC-IV critical-care dataset (Johnson et al., 2023, 2016) re-expressed in the CLIF v2.1 schema, with ∼70,000 ICU stays. Method developers train and self-evaluate against MIMIC-CLIF; because the schema is identical to consortium-side data, the same code runs unmodified when later executed federated. This mirrors the "develop-on-public, validate-on-private" pattern that has become canonical in clinical AI but is rarely operationalized end-to-end.

Consortium cohort

Coverage. Across the 12+ sites currently in CLIF, FLAIR operates on an aggregate cohort of ∼500,000 ICU stays spanning urban academic medical centers, large community hospitals, and health systems with mixed payer populations. Demographic distribution (sex, race, ethnicity, age), full-hospitalization SOFA (worst-in-window), and Charlson and Elixhauser comorbidity indices are summarized per task in the cohort report (`table1.json`). Table 1 reports the MIMIC-CLIF entry-point cohort per task; the consortium-wide pooled Table 1 will replace it at submission with HIPAA-compliant cell suppression applied site by site.

	T1 mort.	**T2 LTACH**	**T3 ext-fail**	**T4 sepsis**	**T5 AKI**
ICU stays	64,434	64,434	26,771	84,854	65,577
Decision rows	198,125	198,125	26,771	7,093,633	4,135,079
Positive rate (row)	19.2%	14.3%	5.4%	3.3%	1.7%
Stays ever positive	13.0%	6.4%	5.4%	54.9%	4.8%
Sex: F / M	27,805/36,629	27,805/36,629	10,033/16,738	37,552/47,302	28,867/36,710
Age 65+ share	55.3%	55.3%	54.1%	54.6%	55.1%
SOFA hosp. max (med., IQR)	5 (2–8)	5 (2–8)	7 (5–10)	5 (2–8)	4 (2–7)
Charlson (median, IQR)	2 (0–4)	2 (0–4)	2 (0–4)	2 (0–4)	2 (0–4)

Table 1: MIMIC-CLIF cohort snapshot per task, generated by `flair extract-metadata -clif-config clif_config_mimic.json` (`flair_benchmark` v2.0.0; HIPAA cell-suppression at $n<10$). *Decision rows* are one-per-stay for T3 (extubation episode) and one-per-timestep otherwise (daily at 07:00 local for T1/T2; hourly for T4/T5). *Stays ever positive* equals the row-level rate for T3 because the task is one-per-stay. *SOFA hosp. max* is the worst SOFA score across the full hospitalization [*admission_dttm*, *discharge_dttm*], computed via `clifpy`'s polars backend (`compute_sofa_polars`). Charlson and Elixhauser are computed from `HospitalDiagnosis`; 49–54% of stays are missing comorbidity scores in the MIMIC-CLIF export, a known data-quality gap (Limitations).

Privacy-preserving access. Three properties together constitute FLAIR's access model. *No patient-level data leaves any site*: outputs are JSON aggregates produced by the FLAIR library inside the container. *Library-enforced suppression*: counts below 10 in cohort tables become `"<10"` in `table1.json`; demographic subgroups with $n < 10$ are dropped entirely from `fairness.json`; calibration bins with fewer than 10 observations report a null event rate in `curves.json`. *Verifiable execution*: a SHA-256 digest of the task script and the FLAIR library version are stamped into every output JSON, so the central aggregator can confirm that the same code ran byte-for-byte at each site. Network isolation is enforced one layer below, by the CLIF-WorkBench container in which the library runs — no internet, no root, immutable read-only image — consistent with the

spirit of analysis platforms such as OpenSAFELY (Collaborative, 2021) in which the code travels and the data does not. Together these properties move evidence generation from the slow, bilateral DUA path to a routine, repeatable operation.

4 Benchmarks

Task taxonomy

Every FLAIR task is declared on three axes in a single `META` dictionary: *task class* (*outcome*, *deterioration*, *intervention*, or *operational*); *temporal pattern* (*episode*, *continuous*, or *fixed horizon*); and *prediction unit* (*one-per-stay*, *one-per-decision-point*, or *one-per-timestep*). The triple determines the shape of the cohort and prediction parquets, which fields are mandatory, and which evaluation primitives the report builder invokes. New task authors fill the triple before writing any SQL or feature code; reviewers consume it before opening a single line of the implementation. The taxonomy is deliberately small so that the boundary between "new task" and "existing task" is unambiguous.

Initial five-task suite

The v1 release covers two outcome targets, one deterioration target, and two intervention-anchored targets spanning ICU operations, kidney, and infectious physiology.

- **T1 — ICU Daily In-Hospital Mortality (07:00 local).** One-per-timestep, 24-hour input window anchored at daily 07:00. Cohort: first ICU stay with discharge in {long-term acute care, mortality}. Label: in-hospital death or transfer to hospice during that ICU stay.
- **T2 — ICU Daily Long-Term Acute Care Discharge (07:00 local).** Complementary label to T1 on the same daily grid: prediction that the patient will be discharged to long-term acute care rather than dying or being discharged home. Together T1 and T2 capture the disposition decision that anchors much downstream ICU planning.
- **T3 — Extubation Failure within 24 h.** One-per-stay, anchored at the first confirmed invasive-mechanical-ventilation extubation event identified by the consensus-window episode detector. Label: death or reintubation within 24 hours. Benchmark targets the classic respiratory liberation question (Thille et al., 2011).
- **T4 — Sepsis (antibiotic-anchored) within 6 h.** One-per-timestep on an hourly grid; t_{zero} is the first qualifying systemic intravenous antibiotic. Cohort follows the Sepsis-3 hospital-wide pattern (Singer et al., 2016; Seymour et al., 2016; Rhee et al., 2019); leakage prevention drops decision points at or after t_{zero}. The implementation mirrors the antibiotic-anchored evaluation framing of Kamran et al. (2024).
- **T5 — Acute Kidney Injury, KDIGO stage 2 within 24 h.** One-per-timestep on an hourly grid; KDIGO stage 2 is defined on serum creatinine with Koyner-style exclusions (Koyner et al., 2018), complementing the broader signature of in-patient AKI prediction (Tomašev et al., 2019).

Each cohort parquet carries a fixed ten-column schema — identifiers, window endpoints, label, split, age, sex, race, and ethnicity — so that downstream feature engineering and evaluation code is shared across tasks.

Leaderboard mechanics

For each task and site, FLAIR reports area under the receiver-operating characteristic curve (AUROC), area under the precision–recall curve (AUPRC), and F_1 at the operating threshold, each with 1,000-iteration bootstrap 95% confidence intervals. Calibration is summarized by slope, intercept, and Brier score, with raw reliability points downsampled into `curves.json` so that the central dashboard can render reliability and ROC/PR curves without re-touching patient data. Fairness is reported as per-subgroup AUROC across sex, race, ethnicity, and age bins together with explicit parity gaps. The leaderboard ranks submissions on the primary metric for each task and displays per-site variance prominently: a single high pooled AUROC hides exactly the failure mode FLAIR exists to surface.

An open, extensible task catalog

FLAIR is released open source under the Apache 2.0 license at `github.com/clif-consortium/FLAIR`, and the five tasks above are explicitly the *seed* of a growing catalog rather than the benchmark's permanent scope. Any researcher or clinician can propose a new task as a single Python module under `flair_benchmark/tasks/` containing the `META` dictionary and a `build` function returning the canonical cohort frame; no registry edits or library changes are required. Acceptance review verifies (i) reproducibility against MIMIC-CLIF, (ii) clinical face validity through a clinician-led review, and (iii) site-governance acceptance at a minimum of two consortium sites before the task enters the federated suite. This is the structural counter to benchmark saturation: the catalog grows with community need, distributed across many proposers, rather than ossifying around the priors of any single author group (Recht et al., 2019; Chekroud et al., 2024).

5 Protocols

The federated protocol is what makes FLAIR's privacy properties operational. We describe it in three steps — developer-side submission, site-side execution, and central aggregation — and then position the protocol against the closest precedents.

Submission

A clinical decision support developer using FLAIR works entirely against MIMIC-CLIF during development. The library exposes CLIF tables through `clifpy`, the canonical CLIF Python interface, and ships a single CLI: `flair load-task` builds a task-specific cohort parquet; the developer engineers features and trains a model on the cohort and the wide CLIF feature set; `flair build-report` emits the four aggregate JSONs. Submission to the consortium consists of code, a frozen model artifact, and a container image specification (Docker or Singularity) that wraps both. The model-to-data paradigm itself is not new — it was introduced by the DREAM Challenges (Guinney and Saez-Rodriguez, 2018) and demonstrated for EHR-based mortality prediction at scale in the EHR DREAM Challenge (Bergquist et al., 2024). FLAIR's contribution is to move it from a one-time competition format into a standing benchmark and submission service.

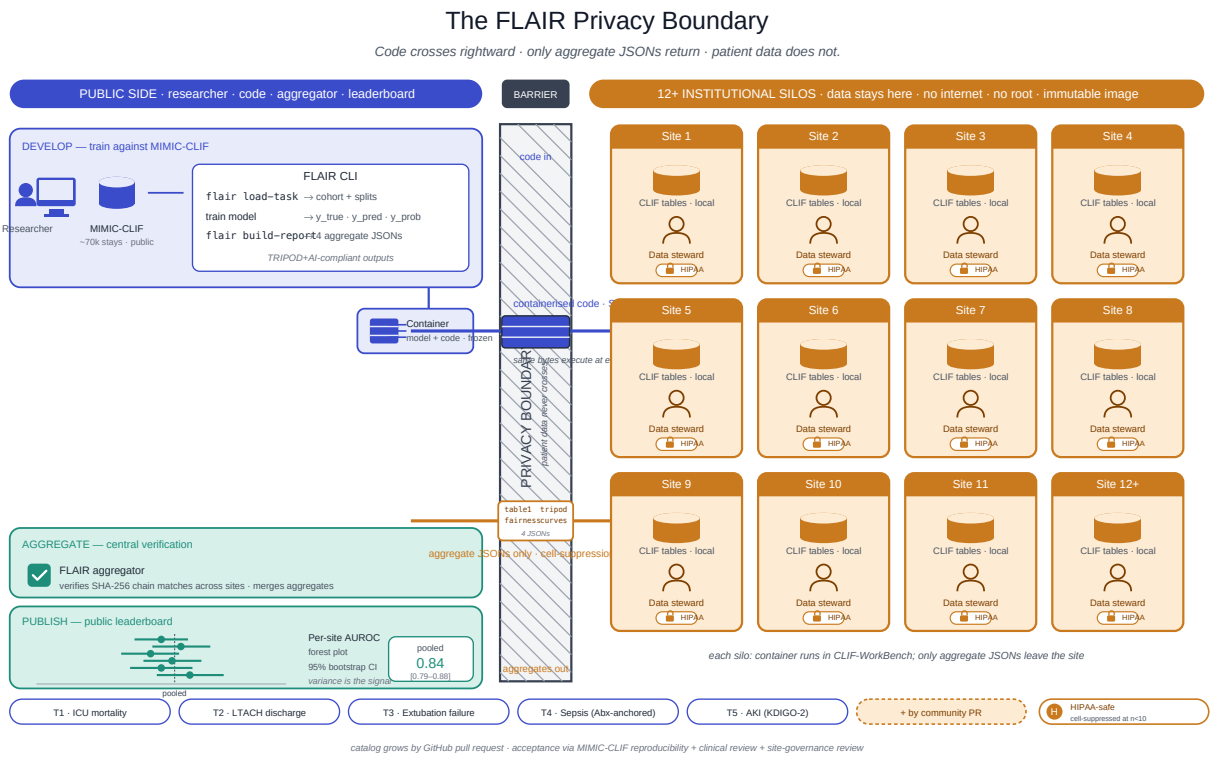


Figure 1: **The FLAIR privacy boundary.** The federation's central contract is visible at a glance: patient-level data never crosses the boundary band, and exactly two things do. *Code crosses rightward*: a developer trains against MIMIC-CLIF using the FLAIR CLI and ships a containerised model whose task script is SHA-256-stamped, so byte-identical execution at every site is verifiable. *Aggregate JSONs cross leftward*: each of the 12+ CLIF silos runs the container inside CLIF-WorkBench (no internet, no root, immutable image) and emits only four reports (`table1.json`, `tripod_ai.json`, `fairness.json`, `curves.json`) with cell suppression at $n < 10$. The FLAIR aggregator verifies the SHA-256 chain across sites and publishes per-site and pooled metrics on the public leaderboard; the per-site forest plot — not the pooled scalar — is the headline signal.

Execution

Each consortium site runs submitted containers inside the CLIF-WorkBench environment: a Singularity container with no internet access, no root, and an immutable read-only image. The submitted code is permitted to read the site's CLIF tables, write a predictions parquet (cohort plus `y_true`, `y_pred`, `y_prob`), and invoke `flair build-report`, which produces exactly four JSON files: `table1.json`, `tripod_ai.json`, `fairness.json`, `curves.json`. Every JSON carries the SHA-256 digest of the task script that produced it, the FLAIR library version, and the generation timestamp. Sites retain final approval over which models execute; the simplicity of the per-task module — one Python file expressing cohort, label, and exclusions in a few hundred lines — makes that pre-execution audit tractable.

Table 2: FLAIR alongside adjacent federated infrastructures.

	OHDSI/LEGEND	MedPerf	FeTS	OpenSAFELY	**FLAIR**
Data model	OMOP	task-specific	imaging (brain MRI)	TPP / EMIS	CLIF
Federation unit	SQL / coefficients	container	training gradients	analytic code	container
Modern DL CDS	no	yes (imaging)	yes (training only)	limited	yes (longitudinal EHR)
PR-extensible tasks	no	partial	no	n/a	yes
Sites (current)	many	~10s	71	English NHS	12+

Aggregation, leaderboard, and lifecycle

The central aggregator verifies that the SHA-256 stamp matches across sites for a given task version, merges aggregate JSONs, and publishes per-site and pooled metrics. Per-site variance is the primary signal of generalization; pooled metrics are reported but de-emphasized. The leaderboard records both retrospective benchmark performance and, for tasks where sites have opted in, prospective silent-trial evaluation under emerging guidance for clinical AI preregistration (Goodman et al., 2025; Joshi et al., 2025). Recent work on operational templates for monitoring AI in routine clinical practice (NEJM AI Pragmatic Trial Authors, 2024) informs the prospective arm.

Comparison with adjacent federated infrastructures. The closest precedent for federated multi-site evaluation in healthcare is the OHDSI network, whose LEGEND framework standardized network studies on the OMOP common data model (Reps et al., 2020); LEGEND federates SQL queries and coefficient sharing for logistic models. MedPerf (Karargyris et al., 2023) demonstrates federated benchmarking for medical imaging models packaged as containers; the FeTS Challenge (Pati et al., 2022) established federated training, rather than validation, as a viable mode for imaging. OpenSAFELY (Collaborative, 2021) delivers a code-travels-not-data analytic platform for national-scale primary-care data. FLAIR contributes the missing combination for critical-care EHR-based CDS: modern longitudinal model classes, a clinically scoped data schema (CLIF) shared across sites, and a community-extensible task catalog under open governance. The five-attribute comparison is in Table 2.

6 Toward the Evidence the FDA Expects from EHR-Based CDS

Recent United States Food and Drug Administration guidance on clinical decision support software (U.S. Food and Drug Administration, 2022), together with the International Medical Device Regulators Forum's framework for the clinical evaluation of software as a medical device (International Medical Device Regulators Forum, 2017), asks developers of EHR-based CDS to make the basis of their recommendations transparent to the clinicians who use them and to demonstrate validation in the populations of intended use. Achieving that posture has historically been bottlenecked by the same access constraints described in the introduction: the relevant evidence is generated, if at all, one bilateral data use agreement at a time (Finlayson et al., 2021; Koch et al., 2024; Soin et al., 2025).

FLAIR's contribution to that posture is structural rather than declarative. Byte-identical task code stamped into every site's output makes the basis of a model's behavior re-inspectable rather than asserted; a reviewer can confirm that the cohort definition, exclusions, and outcome label that produced a site's metrics are the same ones that produced every other site's metrics, without taking that on trust. The per-site discrimination, calibration, and fairness JSONs surface the kind of population-level evidence that regulator-facing dossiers and clinician-facing documentation increasingly require, in a form that is mechanically auditable. And running across 12+ heterogeneous academic medical centers without data movement provides evidence of behavior under realistic distribution shift, where the literature has repeatedly shown that performance can silently degrade (Wong et al., 2021; Finlayson et al., 2021; Koch et al., 2024).

FLAIR is not a regulatory submission, not a substitute for prospective clinical trials, and not a guarantee of clinical utility. It is the layer of evidence between local model development and the regulator- or clinician-facing claims a developer eventually wants to make. That layer is currently missing for most published EHR CDS; making it routine is the goal.

7 Discussion

FLAIR combines three properties that, individually, have appeared elsewhere but together have not been operationalized at the scale ICU CDS development requires: federation across many heterogeneous sites without patient-data movement, a clinically scoped data schema (CLIF) that makes the analytic code portable, and an extension protocol that lets the task catalog grow by community pull request rather than by single-author fiat.

Saturation resistance through siloed data. A side effect of the model-to-data design is that the data underlying FLAIR's leaderboard never enters public hands. Test-set contamination of pretraining corpora — now a recurring problem for open benchmarks (Recht et al., 2019; Chekroud et al., 2024) — is structurally limited, and the incentive for a single developer to overfit the leaderboard is weak because the data they would overfit to are not accessible to them. We view this not as a feature of FLAIR per se but as an emergent property of the access model.

Limitations. Adoption is bounded by the institutions that have implemented CLIF; non-CLIF sites cannot currently participate without an ETL effort. The current task suite is restricted to binary classification on structured EHR variables and excludes notes, waveforms, and images. Aggregate-only outputs constrain some fine-grained bias analyses; subgroups below the suppression threshold are not visible. T5 currently uses serum-creatinine-only KDIGO, and T4's antibiotic vocabulary is an open dependency.

Roadmap. Near-term work extends the suite to opt-in prospective silent-trial evaluation (Joshi et al., 2025), longitudinal calibration drift monitoring against site reference periods, and integration of additional CLIF tables (diagnoses, procedures) as the schema matures. The model-to-data primitive underpinning FLAIR is data-modality-agnostic; once CLIF stabilizes multimodal extensions, the same submission and aggregation contract will apply to notes and waveforms.

Conclusion. FLAIR makes privacy-preserving multi-site external validation of EHR-based clinical decision support a routine, reproducible operation. The library, task definitions, container, and

report generators are open source (Apache 2.0, `github.com/clif-consortium/FLAIR`); researchers and clinicians are invited to add new benchmark tasks through the single-file PR template and two-site governance review described in §4, so the catalog grows with the field.

Required statements

Data and Code Availability. Source code, task definitions, and report-generation library are released under Apache 2.0 at `github.com/clif-consortium/FLAIR`. MIMIC-CLIF is distributed via PhysioNet; consortium data are accessible only by running submitted code at each site through the FLAIR portal. Baseline model weights are deposited on Zenodo (**TODO:** DOI); the CLIF-WorkBench Singularity image is published on GHCR (**TODO:** tag).

IRB. Each consortium site obtained independent institutional review board approval; central protocol coordination is provided by the University of Chicago.

Funding. **TODO:** grants.

Author Contributions. CRediT taxonomy slots **TODO:** fill.

Conflicts of Interest. **TODO:** disclosures.

AI-Use Disclosure. **TODO:** name LLM(s) and the sections or artifacts they helped produce.

References

Roy Adams, Katharine E. Henry, Anirudh Sridharan, Hossein Soleimani, Andong Zhan, Nishi Rawat, Lauren Johnson, David N. Hager, Sara E. Cosgrove, Andrew Markowski, et al. Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis. *Nature Medicine*, 28:1455–1460, 2022. doi: 10.1038/s41591-022-01894-0.

Bert Arnrich, Stephanie L. Hyland, et al. The medical event data standard (MEDS): A common data model for machine learning on electronic health records. *NEJM AI*, 2025. doi: 10.1056/AIra2501253.

Timothy Bergquist, Thomas Schaffter, Yao Yan, Thomas Yu, Justin Prosser, Jifan Gao, Guanhua Chen, Lukasz Charzewski, Zahra Nabian, Ibtihal Husain, et al. Crowdsourcing generalizable mortality prediction models in the EHR DREAM challenge. *Journal of the American Medical Informatics Association*, 2024. doi: 10.1093/jamia/ocad215. PMC10746301.

Rickey E. Carter et al. Multi-site external validation of an artificial-intelligence electrocardiogram algorithm for the detection of reduced left ventricular ejection fraction. *JACC Advances*, 2025. doi: 10.1016/j.jacadv.2025.102537.

Adam M. Chekroud et al. Illusory generalizability of clinical prediction models. *Science*, 2024. doi: 10.1126/science.adg8538.

The OpenSAFELY Collaborative. OpenSAFELY: A platform for analysing electronic health records securely. *Nature Medicine*, 2021. Code-travels-not-data analytic platform for English primary-care EHRs.

Gary S. Collins, Karel G. M. Moons, Paula Dhiman, Richard D. Riley, Andrew L. Beam, Ben Van Calster, Marzyeh Ghassemi, Xiaoxuan Liu, Johannes B. Reitsma, Maarten van Smeden, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. doi: 10.1136/bmj-2023-078378.

Samuel G. Finlayson, Adarsh Subbaswamy, Karandeep Singh, John Bowers, Annabel Kupke, Jonathan Zittrain, Isaac S. Kohane, and Suchi Saria. The clinician and dataset shift in artificial intelligence. *New England Journal of Medicine*, 385(3):283–286, 2021. doi: 10.1056/NEJMc2104626.

Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64 (12):86–92, 2021. doi: 10.1145/3458723.

Steven N. Goodman et al. Preregistration of clinical artificial intelligence studies. *NEJM AI*, 2025. doi: 10.1056/AIe2400146.

Justin Guinney and Julio Saez-Rodriguez. Alternative models for sharing confidential biomedical data. *Nature Biotechnology*, 36:391–392, 2018. doi: 10.1038/nbt.4128.

Anand R. Habib, Anthony L. Lin, and Richard W. Grant. The epic sepsis model falls short — the importance of external validation. *JAMA Internal Medicine*, 181(8):1040–1041, 2021. doi: 10.1001/jamainternmed.2021.3333.

HL7 International. mCODE: Minimal common oncology data elements — HL7 implementation guide. HL7 FHIR Implementation Guide STU4, 2024.

International Medical Device Regulators Forum. Software as a medical device (samd): Clinical evaluation. IMDRF/SaMD WG/N41 FINAL:2017, 2017.

Alistair E. W. Johnson, Tom J. Pollard, Lu Shen, Li-wei H. Lehman, Mengling Feng, Mohammad Ghassemi, Benjamin Moody, Peter Szolovits, Leo Anthony Celi, and Roger G. Mark. MIMIC-III, a freely accessible critical care database. *Scientific Data*, 3:160035, 2016. doi: 10.1038/sdata.2016.35.

Alistair E. W. Johnson, Lucas Bulgarelli, Leo Shen, Alvin Gayles, Ayad Shammout, Steven Horng, Tom J. Pollard, Sicheng Hao, Benjamin Moody, Brian Gow, et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10:1, 2023. doi: 10.1038/s41597-022-01899-x.

Shalmali Joshi et al. Silent trials of clinical artificial intelligence: A scoping review. *Nature Health*, 2025. doi: 10.1038/s44360-025-00048-z.

Fahad Kamran, Donna Tjandra, Andrew Heiler, Jessica Virzi, Karandeep Singh, Jessie E. King, Thomas S. Valley, and Jenna Wiens. Evaluation of sepsis prediction models before onset of treatment. *NEJM AI*, 1(3), 2024. doi: 10.1056/AIoa2300032.

Alexandros Karargyris, Renato Umeton, Micah J. Sheller, Alejandro Aristizabal, Johnu George, Anna Wuest, Sarthak Pati, Hasan Kassem, Maximilian Zenk, Ujjwal Baid, et al. MedPerf: Open benchmarking platform for medical artificial intelligence using federated evaluation. *Nature Machine Intelligence*, 5:799–810, 2023. doi: 10.1038/s42256-023-00652-2.

Bryan Koch et al. A generalizable approach for distribution shift monitoring in clinical machine learning models. *npj Digital Medicine*, 2024. PMC11082139.

Jay L. Koyner, Kyle A. Carey, Dana P. Edelson, and Matthew M. Churpek. The development of a machine learning inpatient acute kidney injury prediction model. *Critical Care Medicine*, 46(7): 1070–1077, 2018. doi: 10.1097/CCM.0000000000003123.

Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, pages 220–229, 2019. doi: 10.1145/3287560.3287596.

NEJM AI Pragmatic Trial Authors. A novel playbook for pragmatic trial operations to monitor and evaluate ambient artificial intelligence in clinical practice. *NEJM AI*, 2024. doi: 10.1056/AIdbp2401267. Manuscript ID AIdbp2401267.

Sarthak Pati, Ujjwal Baid, Brandon Edwards, Micah Sheller, Shih-Han Wang, G. Anthony Reina, Patrick Foley, Alexey Gruzdev, Deepthi Karkada, Christos Davatzikos, et al. Federated learning enables big data for rare cancer boundary detection. *Nature Communications*, 13:7346, 2022. doi: 10.1038/s41467-022-33407-5.

Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. Do ImageNet classifiers generalize to ImageNet? In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 5389–5400, 2019.

Jenna M. Reps, Ross D. Williams, Seng Chan You, Thomas Falconer, Evan Minty, Alison Callahan, Patrick B. Ryan, Rae Woong Park, Hong-Seok Lim, and Peter Rijnbeek. Feasibility and evaluation of a large-scale external validation approach for patient-level prediction in an international data network: Validation of models predicting stroke in female patients newly diagnosed with atrial fibrillation. *BMC Medical Research Methodology*, 20:102, 2020. doi: 10.1186/s12874-020-00991-3.

Chanu Rhee, Zilu Zhang, Sameer S. Kadri, David J. Murphy, Greg S. Martin, Elizabeth Overton, Christopher W. Seymour, Derek C. Angus, Raymund Dantes, Lauren Epstein, et al. Sepsis surveillance using adult sepsis events simplified eSOFA criteria versus Sepsis-3 sequential organ failure assessment criteria. *Critical Care Medicine*, 47(3):307–314, 2019. doi: 10.1097/CCM.0000000000003521.

Juan C. Rojas, Patrick G. Lyons, et al. The Common Longitudinal ICU Data Format (CLIF): A standardized critical care data schema for multi-center research. *Intensive Care Medicine*, 2025. medRxiv 2024.09.04.24313058; consortium spans 9 systems, 39 hospitals, ∼111,000 admissions at founding.

Eliane Röösli, Selen Bozkurt, and Tina Hernandez-Boussard. Bias at warp speed: How AI may contribute to the disparities gap in the time of COVID-19. *Scientific Data*, 2022. doi: 10.1038/s41597-021-01110-7.

Mark P. Sendak, Armando D. Bedoya, et al. Portability of a sepsis risk stratification model across health systems. *npj Digital Medicine*, 2025. Sepsis Watch external evaluation at Summa Health.

Christopher W. Seymour, Vincent X. Liu, Theodore J. Iwashyna, Frank M. Brunkhorst, Thomas D. Rea, Andre Scherag, Gordon Rubenfeld, Jeremy M. Kahn, Manu Shankar-Hari, Mervyn Singer, et al. Assessment of clinical criteria for sepsis: For the third international consensus definitions for sepsis and septic shock (sepsis-3). *JAMA*, 315(8):762–774, 2016. doi: 10.1001/jama.2016.0288.

Mervyn Singer, Clifford S. Deutschman, Christopher W. Seymour, Manu Shankar-Hari, Djillali Annane, Michael Bauer, Rinaldo Bellomo, Gordon R. Bernard, Jean-Daniel Chiche, Craig M. Coopersmith, et al. The third international consensus definitions for sepsis and septic shock (sepsis-3). *JAMA*, 315(8):801–810, 2016. doi: 10.1001/jama.2016.0287.

Arjun Soin et al. Multi-hospital evaluation of strategies to mitigate performance degradation of a clinical prediction model under data shift. *JAMA Network Open*, 2025. PMC12138723.

Arnaud W. Thille, Anatole Harrois, Frédérique Schortgen, Christian Brun-Buisson, and Laurent Brochard. Outcomes of extubation failure in medical intensive care unit patients. *American Journal of Respiratory and Critical Care Medicine*, 183(12):1727–1733, 2011. doi: 10.1164/rccm.201010-1715OC.

Nenad Tomašev, Xavier Glorot, Jack W. Rae, Michal Zielinski, Harry Askham, Andre Saraiva, Anne Mottram, Clemens Meyer, Suman Ravuri, Ivan Protsyuk, et al. A clinically applicable approach to continuous prediction of future acute kidney injury. *Nature*, 572:116–119, 2019. doi: 10.1038/s41586-019-1390-1.

U.S. Food and Drug Administration. Clinical decision support software — guidance for industry and food and drug administration staff. FDA Guidance Document, 2022.

Robin van de Water, Hendrik Nils Aurel Schmidt, Paul Elbers, Patrick Thoral, Bert Arnrich, and Patrick Rockenschaub. Yet another ICU benchmark: A flexible multi-center framework for clinical ML. *arXiv preprint arXiv:2306.05109*, 2023. doi: 10.48550/arXiv.2306.05109.

Andrew Wong, Erkin Otles, John P. Donnelly, Andrew Krumm, Jeffrey McCullough, Olivia DeTroyer-Cooley, Justin Pestrue, Marie Phillips, Judy Konye, Carleen Penoza, et al. External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. *JAMA Internal Medicine*, 181(8):1065–1070, 2021. doi: 10.1001/jamainternmed.2021.2626.

Hugo Yèche, Rita Kuznetsova, Marc Zimmermann, Matthias Hüser, Xinrui Lyu, Martin Faltys, and Gunnar Rätsch. HiRID-ICU-Benchmark — a comprehensive machine learning benchmark on high-resolution ICU data. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.